

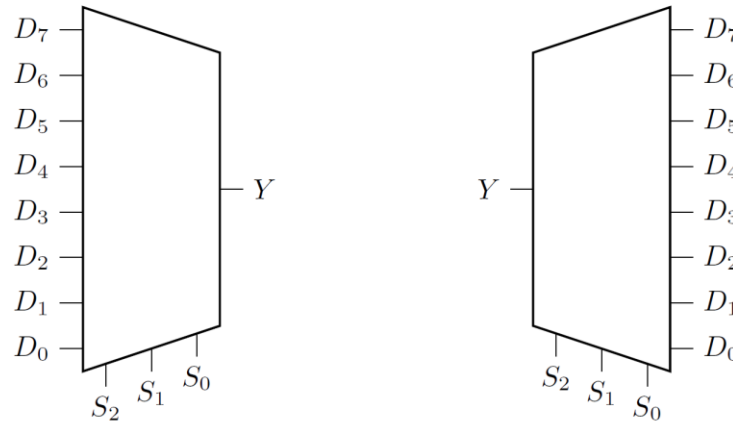
FPGA Programming – VHDL Part

VHDL by Examples – Multiplexer

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Multiplexer

- A Mux selects one signal among 2^n (D_i) thanks to n address lines (S_j)
- A Demux applies the inverse operation



Multiplexer

- 4 to 1 multiplexer in VHDL

entity mux is

```
    port (a, b, c, d: in std_logic;  
          s: in std_logic_vector(1 downto 0);  
          y: out std_logic;
```

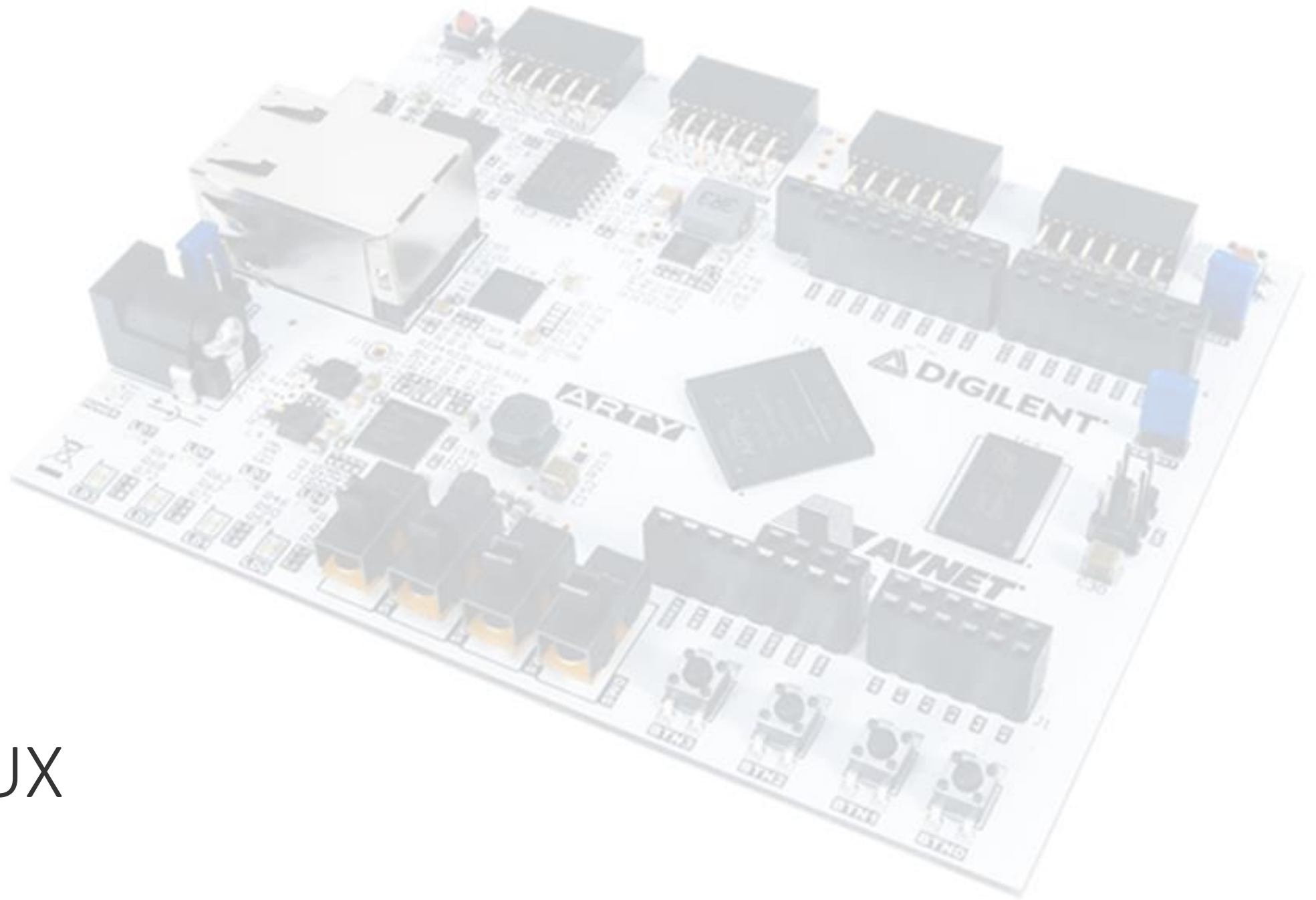
end mux;

architecture Behavioural **of** mux is

begin

```
    y <= a when s = "00" else  
        b when s = "01" else  
        c when s = "10" else  
        d when s = "11" else  
        'X';
```

end architecture Behavioural;



LAB MUX

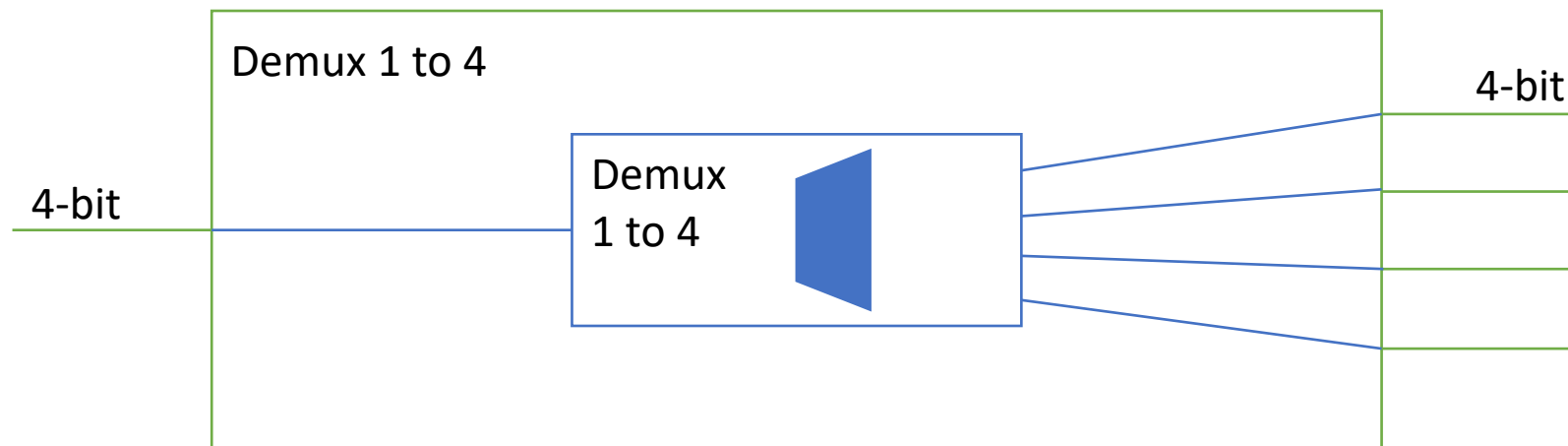
Exercise

- Write a n-bit 1 to 4 demultiplexer of an n-bit input



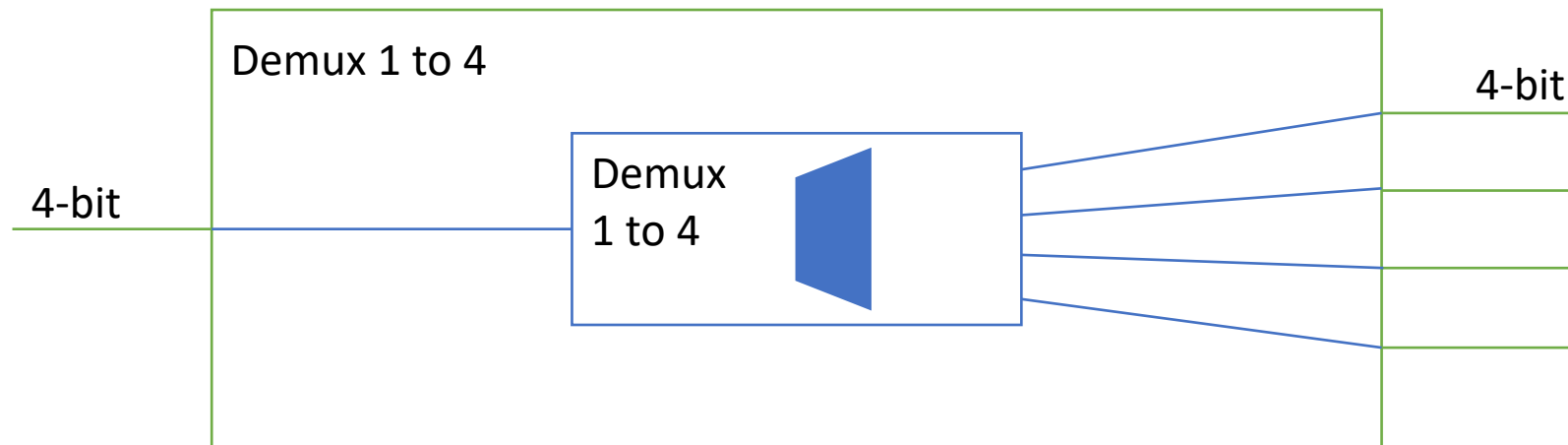
Exercise

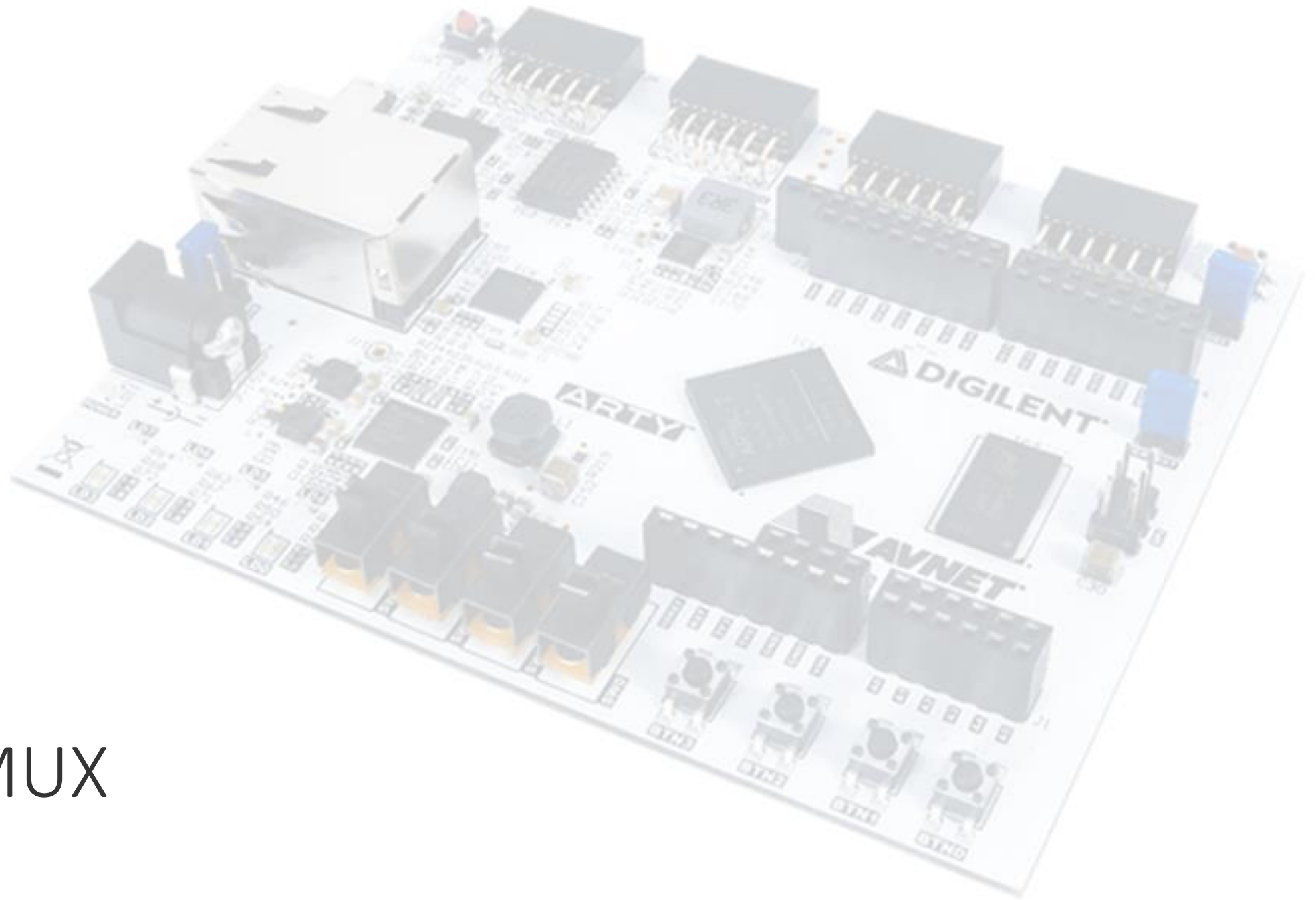
- Write a n-bit 1 to 4 demultiplexer of an n-bit input
- Instantiate it in synthesizable top module (n=4) and simulate it



Exercise

- Write a n-bit 1 to 4 demultiplexer of an n-bit input
- Instantiate it in synthesizable top module (n=4) and simulate it
- Deploy it on Arty A7 board
 - 4 Buttons as input
 - 2 Switches as select
 - 16 Leds as output





EX DEMUX